

Subject Description Form

Subject Code	AAE3008
Subject Title	Fundamental Thermal-fluid Science
Credit Value	3
Level	3
Pre-requisite/ Co-requisite/ Exclusion	Pre-requisite: AP10005 Physics I and AMA2111 Mathematics I
Objectives	1. To provide students with basic knowledge of thermodynamics and fluid mechanics. 2. To develop students' capability of analysing thermal-fluid problems.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Formulate and solve thermal-fluid problems by applying knowledge of thermodynamics and fluid mechanics; b. Analyse and interpret numerical and experimental thermal-fluid predictions and observations; and c. Acquire a basic understanding of the state-of-the-art of this field.
Subject Synopsis/ Indicative Syllabus	Fundamentals of thermodynamics – Basic concepts, including systems, state, processes, equilibrium, thermodynamic cycle; equation of state; First law of thermodynamics, application to closed system; Second law of thermodynamics, concepts of heat engine, refrigerator; Reversible and irreversible systems; Entropy; Entropy of substance. Heat transfer mechanisms – Heat conduction; Convection; Boundary-layer flow; Thermal radiation; Radiative properties; Thermal resistance of parallel walls and cylindrical walls. Kinematic of fluid motion – Properties of fluids; Streamlines, streaklines, and pathlines; Eulerian and Lagrangian descriptions; Material derivative; Reynolds transport theorem; Velocity, vorticity, and strain. Control volume analysis – Mass conservation; Momentum analysis; Energy analysis. Fluid governing equations – Continuity equation, stream function; Momentum equations, stress tensor, constitutive equations, Navier-Stokes equations, Euler equations; Energy equation. Fluid Statics – Fluid pressure; Pascal's law and pressure-height relation; Hydrostatic forces on submerged objects; Buoyancy. Fluid dynamics analysis – Bernoulli's equation; Potential flow, elementary solutions, flow around cylinder, Kutta-Joukowski theorem; Couette flow; Pipe flows – Pipe flow characteristics; Viscosity; Reynolds number; Poiseuille flow; Friction factor; Minor loss.

	External flows – Lift and drag; drag classification, bluff body and streamlined body; boundary layer; flow separation; Drag of flow passing flat plate; Drag of flow around cylinder. Forced heat convection – Nusselt number; Thermal boundary layer, Prandtl number; Force heat convection characteristics for flow passing flat plates, cylinders and spheres; Thermal analysis in circular pipes.				
Teaching/Learning Methodology	Teaching is conducted through class lectures and tutorials. They are aimed at providing students with integrated knowledge required for thermal-fluid applications. Technical/scientific examples and problems are raised and discussed in lecture and tutorial sessions.				
	Teaching/Learning Methodology	Intended subject learning outcomes to be covered			
		a	b	c	
	1. Lecture	✓	✓	✓	
	2. Tutorial	✓	✓	✓	
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed		
			a	b	c
	1. Attendance	5%	✓	✓	✓
	2. Assignment	25%	✓	✓	✓
	3. Test	10%	✓	✓	
	4. Examination	60%	✓	✓	
	Total	100%			
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Overall Assessment: $0.60 \times \text{End of Subject Examination} + 0.40 \times \text{Continuous Assessment}$ The continuous assessment consists of homework assignment and test. They are aimed at evaluating the progress of students’ study, assisting them in self-monitoring of fulfilling the respective subject learning outcomes, and enhancing the integration of the knowledge learnt. The examination is used to assess the knowledge acquired by the students for understanding and analysing the problems critically and independently; as well as to determine the degree of achieving the subject learning outcomes.				

Student Study Effort Expected	Class contact:	
	▪ Lecture	33 Hrs.
	▪ Tutorial	6 Hrs.
	Other student study effort:	
	▪ Self-study	33 Hrs.
	▪ Homework assignment	50 Hrs.
	Total student study effort	122 Hrs.
Reading List and References	1. Cengel Y. A., Cimbala J. M., and Turner R. H., Fundamentals of Thermal-Fluid Sciences. McGraw-Hill, 5th edition. 2. White F. M., Viscous Fluid Flow. McGraw-Hill, 3rd edition. 3. Cengel Y. A. and Ghajar A. J., Heat and Mass Transfer: Fundamentals and Applications. McGraw-Hill, 6th edition.	

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